

Spring Boot REST API Challenge: "Smart Investment Platform"

Scenario

Welcome to **SmartInvest**, a new investment platform.

Customers use a mobile app to:

- View available stocks
- Create investment portfolios
- Buy stocks
- Update portfolio value
- Remove investments

Your team is hired as the backend developer.

Your mission:

Build REST APIs for the SmartInvest investment system using Spring Boot.

Learning Objectives

Students will learn:

- ✓ REST API design
- ✓ HTTP methods
- ✓ URL naming convention
- ✓ Spring Boot Controller
- ✓ Java objects → JSON conversion

Part 1: REST API Detective Challenge (10 minutes)

Scenario

The mobile app needs these functions.

Fill in the missing HTTP method and URL.

User Action	HTTP Method	URL
View all available stocks	?	?
View one stock	?	?
Add new stock into system	?	?
Buy a stock	?	?
Update stock price	?	?
Remove a stock	?	?

Part 2: Create Investment Bean (20 minutes)

Question

Create a Java class:

Investment

with:

- investmentId
- investorName
- stockName
- quantity
- price

and give an example for an object created from the class

Then, convert Investment into Spring Bean. Show the process of creating object from the bean.

Part 3: Create REST Controller (20 minutes)

Question

Create API:

GET /investments

which returns all investments.

What outputs will you get if the following link is tested with a browser?

GET localhost:8080/investments

Part 4: Investment API Bug Hunting 🐛 (10 minutes)

Students become software testers.

Bug 1

Controller:

```
</> Java

@GetMapping("/investment")
public List<Investment> getInvestments(){

}
```

User calls:

```
GET /investments
```

Question:

Why error 404?

Bug 2

Controller:

```
</> Java

@PostMapping("/investments")
public List<Investment> getInvestments(){

}
```

User sends:

```
GET /investments
```

Why error?

Bug 3

Student writes:

```
</> Java

@GetMapping("/stocks")
public String getStocks(){

    return new Stock();

}
```

Why error?